

Complete Project Walkthrough: Indian Corporate Bankruptcy Prediction & Real-Time Explainable AI System

Project Directory: C:\Users\selvraj\.gemini\antigravity\scratch\bankruptcy-prediction
Web Application URL: http://localhost:8501
Primary Model: Multi-Class CatBoost Classifier (99.46% Accuracy, 0.9934 Macro F1)
Explainability Engine: Game-Theoretic TreeSHAP (SHapley Additive exPlanations)
Live Data Feed: Yahoo Finance API (yfinance) across National Stock Exchange (NSE) & Bombay Stock Exchange (BSE)

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1. Executive Summary & Novelty

Predicting corporate distress in emerging economies like India poses unique challenges due to diverse corporate governance, cyclical commodity sectors, high leverage in capital-intensive industries, and the legal framework governed by the **Insolvency and Bankruptcy Code (IBC) 2016** and **Reserve Bank of India (RBI)** prudential norms.

How This System Differs from Existing Academic Systems:

Dimension	Traditional Existing Systems	Our System
Classification Granularity	Binary only (0: Bankrupt, 1: Non-Bankrupt)	4-Tier Granular Classification: Bankrupt , Probably Bankrupt , Financial Distress , Healthy
Data Scope	US / European datasets (e.g. Altman Z-Score Polish data)	Audited Indian filings across 999 Indian firms over a 20-year horizon (2006–2026)
Data Leakage Prevention	Random row-wise train/test splits (temporal & company leakage)	Strict Company-Level Split: 798 train firms, 201 completely unseen test firms (0% leakage)
Explainability	Black-box predictions with no regulator audit trail	Game-Theoretic TreeSHAP (XAI) calculating exact marginal attributions for every metric
Operational Testing	Static historical testing only	Real-Time Live Market Ingestion via Yahoo Finance: evaluates live balance sheets of any NSE/BSE stock
User Experience	Complex command scripts or fragmented multi-page UI	Unified Single-Screen Streamlit Dashboard displaying verdict, ratios, and SHAP on one page



End-to-End System Workflow Architecture

1. Audited Data Feed

8,991 Filings across 999 Indian Firms (2006-2026).
Strict Company-Level Split (798 Train / 201 Test).

2. Model Selection

CatBoost 99.46% (Primary), LightGBM 99.18%, XGBoost 99.13%, Random Forest 98.31%.

3. Live Market Ingestion

Real-Time Yahoo Finance API streaming live balance sheets from NSE/BSE (SUZLON, IDEA, RELIANCE).

4. XAI & UI Dashboard

TreeSHAP marginal factor attributions + Single-screen Streamlit web application.

2. Dataset Overview & Feature Schema

- **Dataset:** Indian Company Financial Dataset.csv
- **Observations:** 8,991 Company-Year records across 999 Indian firms.
- **Audit Time Horizon:** 2006 to 2026 (captures global financial crisis, demonetization, IBC enactment, COVID-19, and recovery).
- **Target Classes (4 Tiers):**
 - 0: Bankrupt (Net worth fully eroded, defaulted on loans, liquidation/NCLT proceedings)
 - 1: Financial Distress (Early-stage operational deterioration, declining interest coverage, margin compression)
 - 2: Probably Bankrupt (Multi-quarter default, recurring operating losses, high leverage exceeding 4.0x)
 - 3: Healthy (Comfortable interest coverage > 2.5x, sustainable leverage, consistent positive operating cash flow)

63 Engineered Financial Indicators:

- 1. **Core Accounting Fundamentals:** Sales, Raw Material Cost, Employee Cost, Other Expenses, Depreciation, Interest, Profit Before Tax (PBT), Tax, Net Profit, Equity Capital, Reserves, Borrowings, Total Liabilities, Total Assets, Receivables, Inventory, Cash & Bank.
- 2. **Cash Flow Fundamentals:** Cash from Operating Activities, Cash from Investing Activities, Cash from Financing Activities, Net Cash Flow.
- 3. **Core Solvency & Profitability Ratios:**
- 4. Interest Coverage Ratio = Operating Profit (EBIT) / Interest Expense
- 5. Debt-to-Equity Ratio = Total Borrowings / (Equity Capital + Reserves)
- 6. Operating Margin = Operating Profit / Total Sales
- 7. Return on Assets (ROA) = Net Profit / Total Assets
- 8. Return on Equity (ROE) = Net Profit / Net Worth
- 9. Working Capital Ratio = Working Capital / Total Assets
- 10. Cash-to-Debt Ratio = Cash & Bank / Total Borrowings
- 11. **Dynamic Multi-Year Growth Metrics:** 1-Year & 3-Year CAGR for Sales, Net Profit, Total Debt, Total Assets, Cash Flow, and historical volatility.

3. Model Training & Cross-Validation Benchmarks

Models were evaluated on **1,839 company-year filings from 201 unseen Indian firms** using a strict company-level split:

Algorithm	4-Class Accuracy	Macro F1-Score	Bankrupt Recall	Distress Recall	Prob. Bankrupt Recall	Healthy Recall
🏆 CatBoost Classifier	99.46%	0.9934	98.9%	98.3%	99.8%	99.8%
🥈 LightGBM Classifier	99.18%	0.9905	100.0%	97.7%	98.5%	99.9%
🥉 XGBoost Classifier	99.13%	0.9898	98.9%	97.1%	99.5%	99.7%
Random Forest	98.31%	0.9720	92.6%	98.6%	95.8%	99.8%
Logistic Regression (Baseline)	74.77%	0.6348	50.5%	27.2%	63.7%	98.3%

 **NOTE:**

CatBoost demonstrated the strongest generalization across all 4 risk tiers with near-zero false alarms and 98.9% detection of bankrupt filings on unseen firms.

4. Explainable AI (TreeSHAP) Architecture

In compliance with banking audits (RBI stress testing) and insolvency court proceedings (NCLT under IBC 2016), black-box predictions are unacceptable. We integrated **TreeSHAP (SHapley Additive exPlanations)** based on cooperative game theory:

TreeSHAP Game-Theoretic Shapley Equation:

$$\phi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \left[\frac{|S|! (|F| - |S| - 1)!}{|F|!} \right] \times [f_x(S \cup \{i\}) - f_x(S)]$$

Top Mathematical Solvency Drivers Across 8,991 Filings:

1. **Interest Coverage Ratio** (Mean |SHAP| = 1.5152): The #1 determinant. Companies unable to cover annual interest from EBIT are overwhelmingly propelled into distress.
2. **Operating Profit Margin** (Mean |SHAP| = 0.5387): Negative operating margins indicate bleeding core cash flows.
3. **Debt-to-Equity Ratio** (Mean |SHAP| = 0.4069): Financial leverage above 3.5x leaves no buffer for macroeconomic or demand downturns.
4. **Return on Assets (ROA)** (Mean |SHAP| = 0.1341): Asset productivity and earnings yield.
5. **Return on Equity (ROE)** (Mean |SHAP| = 0.1331): Capital erosion velocity.

5. Real-Time Live Market Ingestion Engine

Implemented in `live_data_fetcher.py`: * Accepts any ticker symbol (auto-normalizes Indian tickers to `.NS` for National Stock Exchange or `.BO` for Bombay Stock Exchange). * Fetches the latest published Balance Sheet, Income Statement, and Cash Flows from Yahoo Finance (`yfinance`). * Extracts raw statements, computes all 63 ratios and growth trajectories, handles missing items via `indian_imputer.pkl`, and delivers: 1. Live risk verdict and 4-tier confidence percentages. 2. Solvency KPI metrics grid. 3. Real-time TreeSHAP waterfall/bar attribution chart.

6. Simplified Single-Screen Web Dashboard (app.py)

Based on user feedback, the confusing 6-page navigation sidebar has been completely replaced with a **single, unified, uncluttered dashboard**:

UI Layout Structure:

1. **Top Analysis Mode Switcher:**
2.  **Live Stock Ticker (NSE/BSE)**: Test any live listed stock (e.g. `SUZLON.NS`, `IDEA.NS`, `RELIANCE.NS`, `ITC.NS`, `TATASTEEL.NS`, or custom ticker).
3.  **Historical Company**: Instant lookup across 999 firms in the 20-year audited dataset.
4.  **Custom Balance Sheet Numbers**: Interactive sliders/inputs for private company stress-testing.
5. **Unified Results Display (All On The Same Screen):**
6. **Verdict Card**: Large color-coded badge ( **HEALTHY**,  **FINANCIAL DISTRESS**,  **PROBABLY BANKRUPT**,  **BANKRUPT**) with confidence score.
7. **Probability Chart & Ratios Grid**: 4-tier probability distribution bar chart alongside 6 core metrics (Interest Coverage, Debt-to-Equity, Operating Margin, Sales, Net Profit, Debt).
8. **Explainable AI (SHAP) Chart**: Instant horizontal bar chart explaining *why* the AI made this verdict:
 -  **Green bars**: Safe factors keeping the company healthy.
 -  **Red bars**: Distress factors pushing it toward default.
9. **Factor Table**: Side-by-side view of actual reported values vs. exact SHAP marginal impacts.

7. Command-Line Testing Tools

1. Standalone Live Ticker Auditor ([test_live_ticker.py](#))

Query any live stock directly from PowerShell / CMD:

```
# Test Vodafone Idea (Stressed Telecom)
python test_live_ticker.py --ticker IDEA.NS

# Test Suzlon Energy (Turnaround Clean Energy)
python test_live_ticker.py --ticker SUZLON.NS

# Test Reliance Industries (Blue-Chip Conglomerate)
python test_live_ticker.py --ticker RELIANCE.NS

# Interactive Menu
python test_live_ticker.py
```

2. Historical Dataset Tester ([test_model.py](#))

```
python test_model.py
```

Allows searching any company name across the 999 Indian firms (e.g. `Tata` , `Suzlon` , `Adani` , `Alok Industries`).

8. Empirical Validation & Real-World Case Studies

Live external validation on current financial filings:

Case Study A: Suzlon Energy Ltd (SUZLON.NS)

- **Company Status:** Successfully restructured debt, raised equity, and achieved turnaround.
- **Live Financials:** Sales: ₹16,546.7 Cr | Net Profit: ₹3,163.4 Cr | Debt: ₹981.1 Cr | Equity: ₹9,463.5 Cr
- **Ratios:** Interest Coverage = `12.20x` | Debt-to-Equity = `0.10x` | Operating Margin = `16.5%`
- **AI Risk Verdict:** ● HEALTHY (99.89% Confidence)
- **SHAP Drivers:** Interest Coverage (+2.0421), Operating Margin (+0.8390), Conservative Debt-to-Equity (+0.5944).

Case Study B: Vodafone Idea Ltd (IDEA.NS)

- **Company Status:** Severely distressed telecom firm with negative net worth and unsustainable debt.
- **Live Financials:** Sales: ₹44,789.0 Cr | Operating Profit: -₹3,123.0 Cr | Total Debt: ₹192,528.0 Cr | Total Equity: -₹35,758.0 Cr
- **Ratios:** Interest Coverage = `-0.15x` | Debt-to-Equity = `10.00x` | Operating Margin = `-7.0%`
- **AI Risk Verdict:** ● PROBABLY BANKRUPT (99.79% Confidence)
- **SHAP Drivers:** Interest Coverage (+3.2253 towards distress), Debt-to-Equity (+2.4878 towards distress), Negative Operating Margin (+0.3003 towards distress).

Case Study C: ITC Limited (ITC.NS)

- **Company Status:** Cash-rich FMCG blue-chip.
- **Ratios:** Interest Coverage = 300.52x | Debt-to-Equity = 0.03x | Operating Margin = 32.7%
- **AI Risk Verdict:** ● HEALTHY (99.89% Confidence)

9. Step-by-Step How to Run

Prerequisite Dependencies:

```
pip install streamlit plotly pandas numpy joblib shap catboost lightgbm xgboost yfinance
```

Launch the Streamlit Web Application:

```
cd C:\Users\selvraj\.gemini\antigravity\scratch\bankruptcy-prediction
streamlit run app.py
```

Open <http://localhost:8501> in any modern web browser.

10. Repository File Map

```
C:\Users\selvraj\.gemini\antigravity\scratch\bankruptcy-prediction\
├─ app.py # Unified, single-screen Streamlit web application
├─ live_data_fetcher.py # Real-time Yahoo Finance (NSE/BSE) API ingestion & SHAP connector
├─ test_live_ticker.py # Standalone CLI tool to test any live stock ticker
├─ test_model.py # Interactive CLI tester across the 999 historical Indian firms
├─ Indian Company Financial Dataset.csv # Audited dataset (8,991 rows, 999 Indian companies)
├─
├─ models/
│   ├── catboost_indian.pkl # Trained CatBoost model (99.46% accuracy)
│   ├── lightgbm_indian.pkl # Trained LightGBM model (99.18% accuracy)
│   ├── xgboost_indian.pkl # Trained XGBoost model (99.13% accuracy)
│   ├── random_forest_indian.pkl # Trained Random Forest model (98.31% accuracy)
│   ├── indian_imputer.pkl # Median imputer for 63 financial features
│   ├── indian_scaler.pkl # RobustScaler for numerical standardization
│   └─ indian_feature_list.pkl # Canonical ordered list of 63 features
├─
├─ results/
│   ├── confusion_matrices/ # Out-of-sample confusion matrix plots
│   │   ├── cm_indian_catboost.png
│   │   └─ cm_indian_xgboost.png
│   └─ feature_importance/ # Global SHAP beeswarm plots & feature ranking CSV
│       ├── shap_feature_importance.csv
│       ├── shap_bankrupt.png
│       ├── shap_financial_distress.png
│       ├── shap_probably_bankrupt.png
│       └─ shap_healthy.png
```